

Volume III.

§ 95. Equivalent forms and ideals in the quadratic field.

If the two ideals [§ 91, (16)]

$$\mathfrak{a} = (a, a\omega), \quad \mathfrak{a}' = (a', a'\omega') \quad (1)$$

are equivalent, then there is a (integral or non-integral) number η in Ω for which $\eta\mathfrak{a}' = \mathfrak{a}$, and this number η is determined by the ideals $\mathfrak{a}, \mathfrak{a}'$ themselves up to a factor which is a numerical unit. Thus there must be four rational integers $\alpha, \beta, \gamma, \delta$ satisfying the condition

$$\begin{aligned} \eta a' &= a(\alpha - \gamma\omega), \\ \eta a'\omega' &= a(-\beta + \delta\omega). \end{aligned} \quad (2)$$

If, for the moment, we denote by $\eta_1, \omega_1, \omega'_1$ the numbers conjugate to η, ω, ω' , then consequently

$$a'^2 \eta \eta_1 \begin{vmatrix} 1 & \omega' \\ 1 & \omega'_1 \end{vmatrix} = a^2 \begin{vmatrix} \alpha & -\gamma \\ -\beta & \delta \end{vmatrix} \begin{vmatrix} 1 & \omega \\ 1 & \omega_1 \end{vmatrix},$$

that is,

$$a'^2 \eta \eta_1 (\omega' - \omega'_1) = a^2 (\alpha\delta - \beta\gamma) (\omega - \omega_1),$$

and since, by § 91, (12),

$$a(\omega - \omega_1) = a'(\omega' - \omega'_1) = -\sqrt{\Delta},$$

it follows that, if

$$\varepsilon = \alpha\delta - \beta\gamma$$

is put,

$$a\varepsilon = a'\eta\eta_1; \quad (3)$$

and hence, if, as we have already assumed, a and a' are positive and (according to the sharpened notion of equivalence) $\eta\eta_1$ is positive, then ε is also positive.

But if we interchange $\mathfrak{a}$ with $\mathfrak{a}'$, then η passes into $1/\eta$, and ε may pass into ε' . Hence also

$$a'\varepsilon' = \frac{a}{\eta\eta_1}, \quad (4)$$

and consequently $\varepsilon\varepsilon' = 1$. But since ε and ε' are positive rational integers, it follows that $\varepsilon = \varepsilon' = 1$. Therefore, by (2),

$$\omega' = \frac{-\beta + \delta\omega}{\alpha - \gamma\omega}, \quad \omega = \frac{\alpha\omega' + \beta}{\gamma\omega' + \delta}, \quad \alpha\delta - \beta\gamma = 1. \quad (5)$$

Now ω and ω' are the roots of the two forms

$$(a, b, c), \quad (a', b', c') \quad [\text{§ 91, (13)}],$$

and (5) expresses the equivalence of these forms. We therefore have, in agreement with the general theory (§ 88), the theorem:

5. The quadratic forms belonging to equivalent ideals are equivalent.

From this one recognizes the significance of the sharpened notion of equivalence: for if one also wished to allow negative values of $N(\eta)$, then one could also have $\varepsilon = -1$, and the corresponding forms could be improperly equivalent.

Conversely, let $\varphi = (a, b, c)$ and $\varphi' = (a', b', c')$ be two equivalent forms with positive first coefficients a, a' , and let

$$\omega = \frac{-b + \sqrt{\Delta}}{2a}, \quad \omega' = \frac{-b' + \sqrt{\Delta}}{2a'}. \quad (6)$$

Then there is a linear substitution $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ with determinant $+1$ which carries (a, b, c) into (a', b', c') .

Then

$$\begin{aligned} a' &= a\alpha^2 + b\alpha\gamma + c\gamma^2, \\ b' &= 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + 2c\gamma\delta, \\ c' &= a\beta^2 + b\beta\delta + c\delta^2, \end{aligned} \quad (7)$$

$$\omega = \frac{\alpha\omega' + \beta}{\gamma\omega' + \delta}, \quad \omega' = \frac{\delta\omega - \beta}{-\gamma\omega + \alpha}, \quad (8)$$

and a small calculation shows that ω and ω' are corresponding roots, that is, roots with the same sign of $\sqrt{\Delta}$.

Accordingly, for the ideals $\mathfrak{a}, \mathfrak{a}'$ belonging to φ and φ' we obtain two basis forms:

$$\lambda = a(x - \omega y), \quad \lambda' = a'(x' - \omega' y'),$$

and by (8)

$$\lambda = \frac{a[(\delta x - \beta y) + (\gamma x - \alpha y)\omega']}{\gamma\omega' + \delta}.$$

Thus putting

$$x' = \delta x - \beta y, \quad y' = -\gamma x + \alpha y,$$

gives

$$\lambda = \frac{a}{a'(\gamma\omega' + \delta)}\lambda'.$$

Since $a : a'(\gamma\omega' + \delta)$ is a number in Ω , it follows that $\mathfrak{a}$ and $\mathfrak{a}'$ are equivalent.

Thus again in agreement with the general theorem § 88, 4., and at the same time in a more precise determination, we have:

6. If (a, b, c) and (a', b', c') are two equivalent forms with positive first coefficients, and if ω, ω' are corresponding roots of these forms, then the ideals $(a, a\omega)$ and $(a', a'\omega')$ are equivalent.

For negative discriminant the first coefficients of equivalent forms always have the same sign, and one considers only the forms in which this sign is positive. For positive discriminant there are forms with positive first coefficients in every form class, and therefore form classes and ideal classes correspond to one another in a one-to-one way.

A basis of the principal ideal η is $(\eta, \eta\theta)$ and

$$\begin{aligned} \theta &= \frac{\sqrt{\Delta}}{2}, & \theta^2 &= \frac{\Delta}{4}, & \Delta &\equiv 0 \pmod{4}, \\ \theta &= \frac{-1 + \sqrt{\Delta}}{2}, & \theta^2 &= \frac{\Delta - 1}{4} - \theta, & \Delta &\equiv 1 \pmod{4}. \end{aligned}$$

If x, y are rational integers and

$$\eta = x + y\theta,$$

then this basis is, in the two cases,

$$\begin{aligned} (\alpha_1, \alpha_2) &= \left(x + y\theta, \frac{\Delta y}{4} + \theta x \right), \\ &= \left(x + y\theta, \frac{y(\Delta - 1)}{4} + (x - y)\theta \right), \end{aligned}$$

and the substitution determinant A [§ 87, (1)] is

$$x^2 - \frac{\Delta}{4}y^2, \quad x^2 - xy + \frac{1 - \Delta}{4}y^2 = N(\eta).$$

The basis (α_1, α_2) is therefore positive if the norm of η is positive (otherwise α_1 would have to be interchanged with α_2). From the basis form

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 = \eta(t_1 + \theta t_2)$$

one obtains

$$N(\lambda) = N(\eta)T,$$

and accordingly T becomes the principal form

$$\left(1, 0, \frac{\Delta}{4}\right), \quad \left(1, -1, \frac{1-\Delta}{4}\right). \quad (9)$$

If $N(\eta)$ were negative, one would have obtained for T the forms

$$\left(-1, 0, -\frac{\Delta}{4}\right), \quad \left(-1, 1, -\frac{1-\Delta}{4}\right),$$

and these are equivalent to (9) only if Pell's equation

$$t^2 - \Delta u^2 = -4$$

is solvable.